# THE CO-UTTERANCE HYPOTHESIS. AN ACCOUNT OF SIMULTANEITY IN SIGN LANGUAGES

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**Abstract**

In sign languages, different articulators (i.e., the two hands) can represent different characters involved in the same action. As skilled bimodal bilinguals can simultaneously utter a sentence in a sign language and in a spoken language, the simultaneous production of two sentences is cognitively possible. Since in sign languages multiple articulators work with some degree of autonomy, we propose that it is possible to co-utter two or more coordinated clausal units internal to the same sign language if they convey different aspects of the same complex conceptual message. Not only can two propositional units be co-uttered, but also two constituents internal to the same sentence can be. This allows us to develop a new approach to Role Shift in sign languages and related phenomena. We discuss the consequences for theories of linearization, comparing those which claim that syntactic structures are inherently provided with a linear order (based on asymmetric c-command configurations) and those that assume that order is imposed only when syntactic structures are shipped to the phonological component.

**Keywords**

Linearization, Role Shift, Simultaneity, Sign Languages, Incorporation

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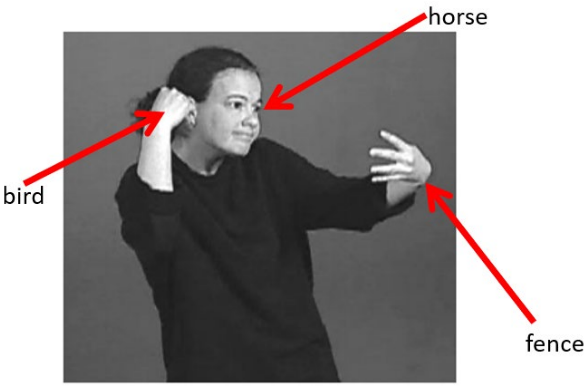
## AN ACCOUNT OF SIMULTANEITY IN SIGN LANGUAGES

### 1. Introduction

More than fifty years of research on sign languages have been marked by a tension between two different approaches. On the one hand, a large body of research has sought to establish that sign languages have the same underlying characteristics of spoken languages, i.e. they have the same expressive power, show "duality of patterning" (namely the property that the meaningful units of language, say words or morphemes, are made up of meaningless units, say phonemes) and display a complex (recursive) syntactic structure. Just for convenience, we will call this 'the unifying approach', as it presupposes that both sign and spoken languages are the byproduct of the same underlying language faculty. According to the unifying approach, notions like phoneme, syllable, morpheme, agreement, etc. are fundamental tools to analyze sign languages and these display the complex syntactic constructions extensively explored in spoken language. Furthermore, sign languages have been shown to vary along dimensions of variations that are familiar from the typological literature, for example relativization may involve internally or externally headed construction as well as correlatives. A typical output of the unifying approach are the descriptive grammars of sign languages that have been developed in recent years (cf. https://thesignhub.eu/grammars).

As rich and successful as this program was, there are aspects of sign languages that are particularly recalcitrant for the unifying approach. A particularly challenging case is highly simultaneous structures that can be illustrated with a representative example taken from Sallandre (2007). The picture in (1) illustrates an utterance that can be expressed in English as follows: 'Here is the bird watching the horse getting ready to jump over the fence'. In (1) eye gaze, facial expression, mouth gestures and body posture in general represent a horse. The dominant hand represents a bird looking at the horse. A fence is expressed by the non-dominant hand. Of course, this utterance cannot come out of the blue but must be part of a narration in which the relevant characters have already been introduced. Still, this massive simultaneity of information poses a challenge for traditional grammatical descriptions. Is this utterance a sentence? Is it a coordination of different clauses? What is its subject? What is the predicate? Can traditional grammatical categories be fruitfully applied to cases like (1)?

(1)




(Picture from Sallandre 2007)

Highly simultaneous structures like (1) in which *three* set of articulators personify three entities at the same time are rather uncommon in normal discourse and require skills that few signers completely control, so one might be tempted to say that they are pantomimic uses embedded in a linguistic utterance. However, the simultaneous use of two articulators to personify two different entities is quite common (cf. Sallandre 2014 for other examples in which three articulators represent three different characters and Slonimska, Özyürek and Capirci 2020 for experimental findings about simultaneous encoding of different interacting referents in LIS). Therefore, leaving highly simultaneous structures outside the linguistic analysis would be neglecting an important feature of sign languages.

In contrast to the unifying approach, the modal approach (cf. Cuxac 2000) posits that modality is a generator of constraints and potentials that determines both the underlying and the superficial form of linguistic expressions. In this approach, the massive presence of iconicity and simultaneity in sign languages can be explained by the potentialities of visual-spatial modality, which allow (i) showing while saying (i.e., iconicity) and (ii) articulating several parts of the body at the same time to convey meanings (i.e., simultaneity). As a result, linguistic expressions carried by the one modality can hardly be transferred to the other modality. This implies that analytical tools from spoken language linguistics cannot be used (or, at minimum, must be massively modified) to be useful for sign languages.

For sure, this view of a field divided between the unifying approach and the modal approach is somewhat simplified, as many researchers in one tradition tried to integrate elements of the other tradition. At the level of semantic analysis, a very productive approach tried to show how meanings units produced by compositional rules of formal semantics interact with iconic representations made available by the visuo-spatial modality (cf. Schlenker 2022 for a general introduction to this approach). However, the divide between the unifying approach and the modal approach is real, as the agenda is very different in the two cases, namely looking for unification in one case and starting from sign language singularities in the other case.

More specifically, how to analyze the internal syntax of 'high simultaneity utterances' like (1) with the toolbox of formal syntax has remained elusive. In this paper, we will adopt the unifying approach and explain highly simultaneous structures as cases of co-utterance of two entire clausal constituents (or two constituents inside the same clause). Co-utterance has been largely neglected, at least in the unifying approach, given that the research overwhelmingly focused on spoken languages in which it is impossible for trivial articulatory reasons. On the other hand, co-utterance of two linguistic categories often takes place in sign languages since two articulators (say the two hands, or one hand and another body part) can work autonomously, although in coordination. For concreteness, we will call ‘co-utterance hypothesis’ the claim that complex cases of simultaneity are cases of simultaneous articulation of categories that are otherwise distinct until the moment of spell-out.

Admittedly, the idea that one can simultaneously utter two full sentences or two categories inside the same sentence (say the subject and the predicate or the verb and its complement) may seem bizarre. To be sure, it is well attested that skilled interpreters can handle two grammatical systems, one in comprehension and one in production (cf. Chernov 1994). However, we are going to assume that it is possible to handle two distinct grammatical systems *in production*. To support this unorthodox view, we will start this paper by offering a proof of concept. Previous literature shows that the simultaneous utterance is massively attested in a specific situation, namely bimodal production by CODAs (Children of Deaf Adults) or KODAs (Kids of Deaf Adults). CODAs and KODAs do blend sign and spoken language, typically when their interlocutors include signers and non-signers. This shows that the simultaneous utterance of sentences, clauses or other syntactic constituents is cognitively possible in two parallel languages. Our move in this paper is arguing that this type of simultaneous production can take place *also internally to a single sign language*. This will allow us to interpret at this light highly simultaneous structures and to analyze them with familiar (yet to be adapted) linguistic categories, by keeping to the unifying approach.

As simultaneity is massive in any sign language component (lexicon, phonology, morphology, syntax and semantics) a single paper cannot deal with all components where simultaneity surfaces. Our paper deals with cases in which two clausal units (to be better defined below) or two constituents within the same sentence are produced at the same time by different articulators instated of being linearized one after another, as is obligatory in spoken languages. More specifically, the construction that we will consider are:

(i) Co-utterance of clausal units each with a separate meaning contributing to the description of a complete complex event. We have already seen an example of this in (1). Others will be considered below.

(ii) Reciprocal sentences in which the two hands represent two entities (or class of entities) that interact.

(iii) Sentences in which the verb is co-uttered either with its subject or with its internal argument. These are cases that go under the name of Role Shift in the sign language literature and are not normally analyzed as cases of co-utterance in this literature.

This paper is organized as follows. Section 2 contains the proof of concept as it briefly reviews the literature on bimodal production in which the simultaneous production of two clauses and of two constituents is reported. In Section 3, we address cases of simultaneity of clausal units like (1), as well as reciprocals sentences. We take them to be cases of simultaneous utterance of two *v*Ps, which are embedded under a unique middle field. In Section 4, we analyze the well-known phenomenon of Role Shift (more specifically, Action Role Shift) as simultaneous production of subject and verb or of object and verb. Section 5 compares the co-utterance hypothesis defended in this paper and an analysis in terms of incorporation, which might seem superficially similar but is quite different. Section 6 discusses some general constraints on co-utterance and the consequences of our approach for the theory that explains how hierarchical structure is flattened in a liner sequence of words/signs. Section 7 concludes the paper.

Before we start, two provisos. The first one is that, although the phenomena we focus on in this paper are shared by many sign languages, we mostly exemplify them with LSF (langue des signes française, French Sign Language). When necessary, we will use examples from other sign languages, though. The judgments that we report are by the Deaf co-author of this paper, for whom LSF the main communication language. He checked his own judgements with two Deaf people with a similar linguistic profile (they are deaf from birth, LSF is their main communication language, and it has been acquired before the age of 3). There was no formal elicitation setting, as the linguistic examples we report came out during of the discussion between the authors of this paper. We do not assume that the linguistic descriptions that we report hold for the entire community of LSF users, which includes a large number of people exposed to sign language relatively late in life, as common with signing communities. However, many of the core phenomena we deal with have been repeatedly reported in the previous literature and are attested in other sign languages, as we will indicate case by case. So, we are confident that our description is reliable.

The second proviso is that the glosses we use in the text cannot do justice to the richness of the sign language utterances that we report. The glosses are meant to express only the aspects that are relevant for the properties discussed at that point of the paper. However, we made available all the videos at the following OSF address and we strongly invite the reader to consult them: https://osf.io/ygjxs?view_only=56436b308b8a4eaeb1e7b635586ec13b. When not differently indicated, sign language examples are LSF sentences whose videos are available on the OSF platform.

## 2. Simultaneous utterance in bimodal production

There is a fairly rich literature on bimodal production by CODAs or KODAs, namely hearing children of Deaf parents who are natively exposed to a sign language within the family and are exposed to the dominant spoken language almost as early from the larger hearing community. We cannot make justice to this literature but focus only on some features that are relevant for our discussion in this paper. Following the terminology proposed by Emmorey at al. (2008), we define 'code blending' the simultaneous production of signs and words, which is possible given the availability of two independent articulatory channels. As in CODAs code blending is more common than code mixing (where signs and words alternate), Emmorey et al. proposed that, when two languages are activated at the same time, the inhibitory process that leads to suppression of one channel over the other is costlier than the simultaneous production of signs and words. Therefore, if articulatory constraints do not force inhibition, the output in the two channels is simultaneously produced.

Baker & van den Bogaerde (2008) propose a classification of code blending and observe that, in addition to cases where the utterance is primarily spoken with occasional concurrent production of signs and cases where the utterance is primarily signed with some accompanying speech, two more substantial cases of code blending must be identified, namely cases in which an entire sentence is fully expressed in both modalities and cases in which a complete utterance results only by adding the contribution of the two modalities (cf. Lillo-Martin et al. 2016 for a systematic description of code blending and Giustolisi et al. 2024.

for an experimental investigation of code blending in LSF). We illustrate the two substantial cases of code blending with an NGT-Dutch production in (2) and with a LIS-Italian production in (3).

(2) NGT: ALL CANNOT

Dutch: allemaal kan niet

all cannot

'None of us can do (that).' (NGT-Dutch, adapted from Baker & van den Bogaerde 2008: 9)

(3) LIS: TALK HUNTER

Italian: parl-a con Biancaneve

talk.3sg with Snow-White

'The hunter talks with Snow White.'

(LIS-Italian, adapted from Branchini and Donati 2016)

(2) is a case in which a complete NGT and a complete Dutch sentence are uttered simultaneously while in (3) Italian and LIS are both needed to complete the sentence: the inflected verb is simultaneously expressed in both languages, but the indirect object is provided only by Italian and the (postverbal) subject is provided only by LIS. These two examples show two points that will be crucial for the discussion below. If different articulators are involved:

(i) two full sentences expressing an identical or very similar meaning can be uttered at the same time (cf. 2)

(ii) two different constituents can be simultaneously expressed (this is the case of the subject, expressed in LIS by the sign HUNTER, which is co-articulated with the indirect object, expressed by the Italian prepositional phrase 'con Biancaneve', in 3).

It is worth stressing that code blending involving two full sentences is possible even when the word/sign order is very different in the relevant spoken and sign language. Italian/LIS is a particularly clear case in this respect as these languages are respectively head initial and head final. This creates cases in which the utterances in the two languages should display different word order, say SVO and SVO. As discussed by Branchini and Donati (2016), these cases of conflict are resolved in different ways: the Italian word order may surface, the LIS word order may surface or two simultaneous utterances with diverging order are produced. Branchini and Donati also note that code blending is not completely costless. In particular, they note that, if language A and B have two different word order rules and language A (LIS) imposes its word order on language B (Italian), there is a morphological and prosodic simplification (deaf voice) in the Italian production.

Code blending shows that that the simultaneous utterance of sentences, clauses or other syntactic constituents is possible. While (2) and (3) involve two different languages, our main move in this paper is proposing that the counterpart of code blending illustrated by (2) and (3) may occur within a single language when the simultaneous expression is not blocked by articulatory constraints. What we are going to propose is in the same spirit of Emmorey's et al. (2008) account to explain cases of code blending like the one below elicited by a cartoon of Sylvester and Tweety. Emmorey et al. propose that the preverbal message contains the event concept (e.g. Tweety suddenly looking at Sylvester), but different parts of the concept are sent to the two languages: the temporal aspect of the event (i.e. the fact that it is sudden) is encoded in English, while the action (looking at someone) is encoded in ASL. This happens because the two languages can efficiently express the two aspects of the preverbal message simultaneously.

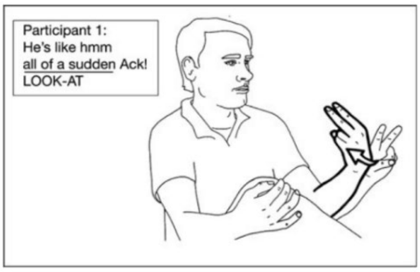



Figure 6. Example of a semantically non-equivalent code-blend. In this example, the participant's body represents the object of LOOK-AT (i.e., Sylvester). ASL illustration copyright © Karen Emmorey.

What we are adding is that two aspects of the preverbal message can be sent separately to the *same* language if this allows efficient simultaneous expression.

More specifically, in the next Section we discuss cases in which two (or more) independent clauses contribute to form a sentential coordination (the monolingual counterpart of 2). In Section 4, we will switch to cases in which two constituents within the same clause are uttered at the same time (the monolingual counterpart of 3 and of the Sylvester and Tweety's example).

### 3. Simultaneous utterance of two clausal constituents

In this section we consider some candidate cases of simultaneous expression of two clausal constituents within the same sign language.

#### 3.1 Simultaneous expression of reciprocity

As already mentioned, in sign languages the simultaneous use of the two hands can serve the function of personifying two different entities and their interaction. One of these cases involves reciprocity. We will focus on LIS but a similar pattern is attested in other sign languages. In LIS there are three productive strategies to express reciprocity. (4) contains a reciprocal marker (glossed as EACH_OTHER). This marker is used with plain verbs.[1] As for agreeing verbs, two strategies to express reciprocity are attested. The first strategy, illustrated in (5), is the sequential reduplication of the verb. The sign for the verb (realized with two hands) starts from the position in the neutral space associated with referent A and moves towards the position associated with referent B and then back from position B to position A. This is a case of reduplication of the verb stem which has been described in spoken as well as in sign languages different from LIS (cf. Pfau and Steinbach 2003 for a survey of reciprocity across modalities and Pfau and Steinbach 2016 for an extension to other types of plurality of relations). However, the case that concerns us more directly is (6), which is discussed by

[1] Plain verbs are negatively defined as those that cannot be spatially modified to agree with their arguments, typically because they are anchored to the body. Agreeing (also called directional) verbs are spatially modified to mark their arguments. Cf. Padden (1988) for a presentation of this typology.

Fornasiero (2020). After the dominant hand has articulated the plural pronominal subject, there is *simultaneous* reduplication of the verb: the two hands move with the handshape and movement of the lexical sign HELP, but with opposite orientation and direction.

(4) IX$_{1+2}$ UNDERSTAND EACH_OTHER

'You and I understand each other.'

(5) IX$_{3a+3b}$ $_{a}$HELP$_{b}$ $_{b}$HELP$_{a}$

'They help each other'

(6) dominant: IX$_{3a+3b}$ $_{a}$HELP$_{b}$

non-dominant: $_{b}$HELP$_{a}$

'They help each other'

(LIS, images from Fornasiero 2020)

We propose to analyze (6) as a case in which, following a hanging topic expressed by the dominant hand, the two hands simultaneously articulate two units with the meanings 'A helps B' and 'B helps A'. We take this to be initial evidence in favour of the co-utterance hypothesis: two articulators can express two clausal units at the same time also inside the same language and not only in case of code blending (we qualify what we mean by clausal units in the next section) .

The two clausal units co-uttered in (6) have a very close meaning. In fact, they have the same thematic structure but for the switching of the internal and external argument, which is reflected in the agreement pattern of the verb. One can ask: is identity of meaning necessary for two sentences to be articulated simultaneously? The question is legitimate since in the cases of code blending involving two full sentences discussed in the literature, as well as in (6), the two sentences are a close translation of each other. However, in the next section we propose that identity of meaning is not necessary.

### 3.2 Simultaneous expression of the interaction between multiple entities

As already hinted at, we want to push the co-utterance hypothesis to cover complex cases like (1). A first attempt is proposing that the utterance in (1) is the simultaneous expression of three independent coordinate clauses:

(i) 'there is a fence',

(ii) 'the horse looks at *X* (*X* being somewhat interpreted as the fence)' and

(iii) 'the bird is watching *Y* (*Y* being somewhat interpreted as the horse)'.

In Section 3.3 we will qualify this claim by saying that what is coordinated is not full clauses but smaller propositional units (*v*Ps). But for time being, let us stick to the hypothesis that what is the co-uttered is clauses.

If we go this way, the constraint that the co-uttered units must express the same meaning must be relaxed, as these units have different content. Furthermore, proposing that there are three independent coordinate clauses is not sufficient: we need to get the complex meaning 'here is the bird that watches the horse that is getting ready to jump over the fence'. In order to achieve that meaning, an extra step is needed, namely making sure that the horse that the bird is looking at is the one that is looking at the fence. This can be done in English by a relativization structure, involving subordination (as opposed to coordination of independent clause). In English, an alternative to relativization would be anaphorically linking the three entities in the three consecutive sentences as in

(i) 'there is a fence'

(ii) 'the horse looks at *it*'

(iii) 'the bird is watching *him*'

where 'it' refers to the fence and 'him' refers to the horse. However, anaphoric linking is less effective than relativization in English because the reference of the pronoun is not determined rigidly. Depending on the context, 'it' may refer to the fence or to something else and similarly for 'him'. Importantly, anaphoric linking is different in sign languages. Instead of a trichotomy between first, second and third person, in sign languages an antecedent is associated with a position or 'locus' in signing space, and an anaphoric link is obtained by pointing towards that locus. In fact, each argument can be assigned a locus and one can unambiguously refer back to that argument by pointing towards that locus, although actual pointing is not necessary (eye gaze to the locus can be sufficient for example).

We propose that the utterance in (1) conveys its complex meaning by a mechanism of *spatial* anaphoric linking (cf. Schlenker et al. 2023 for a formal account of cross-sentential anaphora that adapts to sign language the semantics proposed for pictorial representations): the dominant hand with the bird handshape is oriented toward the locus of the horse and the eye gaze expressing the horse is directed toward the locus where the fence is placed. This makes sure that the three clauses mean that the bird looks at the horse (as opposed to something else) and the horse is looking at the fence (as opposed to something else). So, the complex meaning can be effectively expressed via anaphora but without relativization in this specific case (although in other cases relativization strategies are attested also in LSF, cf. Hauser et al. 2021).

Notice that, if (1) is the simultaneous production of three anaphorically linked clauses, it should be possible to produce these clauses also one after the other, since simultaneity is an option made available by the presence of independent articulators but is not compulsory. In fact, this is correct. The simultaneous and non-simultaneous versions are reported below (7a is a video that reproduces the utterance illustrated by the picture in 1). In both sentences, the anaphoric link is expressed spatially:

(7a)

dominant: BIRD

non-dominant: COW

eye-gaze: FENCE

____LOOK

(7b) FENCE COW LOOK-AT BIRD FLYING

To summarize: if we are on the right track, as far as (1) is concerned, two factors make possible the simultaneous expression of the complex meaning it conveys: the relative independence of three set of articulators (dominant hand, non-dominant hand and eye gaze) and spatial anaphora, which is inherently simultaneous and allows a tight link between three clauses that are otherwise syntactically independent.

### 3.3 Semantic and structural constraints on highly simultaneous structures

By applying the co-utterance hypothesis to cases like (1) we need to relax the constraint that co-uttered sentences must have the same (or very similar) meaning. While this constraint is respected in cases of reciprocals (cf. 6), it is clearly not operative in (1), as the three coordinated clauses give a different meaning contribution to the complex scenario involving the three entities. By doing this, aren't we opening a Pandora's box?  After all, it is never the case that a set of articulators express Pythagoras' theorem and a second set of articulator conveys a comment on yesterday weather! We clearly need to impose limits on the type of contents that can be co-articulated.

However, it is not difficult to see how language production models might explain the semantic constraints on co-articulation of full clauses. For concreteness, we take Levelt's (1989) language production model, but a similar reasoning can be replicated for other approaches (see Ferreira 2010 for a review on sentence production models). Levelt postulates a level responsible for generating the conceptual message (the Conceptualizer), which then transmits this content to a component that linguistically encodes it, the Formulator. Assuming this, we suggest that in order for two sentences to be co-articulated they must stem from the same message at the Conceptualizer level. This easily applies to cases like (6) as the two sentences before the Formulator level, where lexicalization takes place, are identical ('x helps y', *x* and *y* being variables to be lexically filled). The situation is less straightforward for more complex case like (1) but one can imagine that at the Conceptualizer level a message describing the interaction between three entities is generated and at the Formulator level this message is linguistically encoded in three different clauses to be uttered simultaneously.

Another important issue concerns the syntax of co-articulated clauses. The analysis we offered for the cases considered up to now is in terms of coordination. Even when a possible translation in English involves subordination (say relative clause stacking as in 'here is **the bird that** watches **the horse that** is getting ready to jump over the fence'), we argued that that complex meaning is expressed in LSF through coordination and through a spatial anaphoric.

In fact, there are reasons to think that highly simultaneous structures might be cases of coordination of propositional units smaller than a complete clause. The evidence comes from the fact that the referents in highly simultaneous structures are understood to exist at the same time, whereas in consecutive constructions this constraint does not hold. This can be illustrated by the following minimal pair. (8) illustrates a situation in which someone is fired by a handgun. The two hands simultaneously express the firing action and the victim's falling. (9) is minimally different because the gun is a cannon. In this case the two hands cannot simultaneously express the firing action and the victim's falling, plausibly because in real life a temporal lag is perceived between the two events. The only grammatical sentence is the one in which the two hands represent the action consecutively (.

(8)

| | |
|---|---|
| dominant: | GUN |
| non-dominant: | CL_PERS_FALLING |

(9)

| | | |
|---|---|---|
| dominant: | CANNON | |
| non-dominant: | | CL_PERS_FALLING |

It appears that time lag in the real world implies sequentiality in the signing production. Why? Notice that invoking an iconic mapping would not be enough, because the inverse implication does not hold: sequentiality in the signing production does not imply the presence of a time lag in the real word (for example, the sequential counterpart of 1, namely 7b, represents events that are simultaneous in the real world). We propose that the constraint that the events expressed in highly simultaneous structures must occur at the same time in the real world can be given a structural explanation under the co-utterance hypothesis if it is assumed that the co-uttered units are coordinated *v*Ps and not full TPs (we take *v*Ps to be the units formed by the verb and the arguments it assigns a theta-role to). If so, multiple *v*Ps are embedded under the same functional layer expressing tense (and aspect). This is why the same temporal collocation is imposed onto the events expressed by the co-uttered *v*Ps.

That there is a unique functional layer above the two *v*Ps is confirmed by data coming from negation and from the aspectual markers. The simultaneous negative sentence (9) is true when someone is neither reading nor calling, when they are reading but not calling and when they are calling but not reading. These truth conditions are expected if the negative sign in the high part of the functional structure applies to the coordination of the two *v*Ps '*x* phones' and '*x* reads'.

(10)

| | | |
|---|---|---|
| | | _hs |
| dominant: | PHONE | NOT |
| non-dominant: | READ | |

Converging evidence comes from the aspectual information: the events referred to by the two *v*Ps can be both perfective or both imperfective but it cannot be the case that one is perfective and the other one imperfective. For example, the perfective marker DONE cannot be attached to only one of the two *v*Ps: the sentence in (11) cannot mean that the phone call is finished but the reading activity continues.

(11)

dominant: PHONE DONE

non-dominant: READ

Similarly with modal verbs. (12) means that one can read and make a phone call at the same time.

(12)

dominant: PHONE CAN

non-dominant: READ

Finally, highly simultaneous structures are compatible with interrogative elements, showing that the left periphery is projected, but the *wh*-sign is across-the-board extracted from the two simultaneous VPs and this further suggests that there is a unique functional structure above the two *v*Ps.

(13)

dominant: PHONE WHO

non-dominant: READ

### 3.4 Interim Summary

In this section, we have seen cases of the co-uttered 'sentences' stemming from the same Conceptualizer that are complete propositional units (technically, *v*Ps). The fact that they are smaller than full clauses (TPs) must be further explored but it is likely due to the cognitive load that handling two complete structures would imply. In fact, this constraint is reminiscent of the observation concerning code blending introduced in Section 2: in complex cases of code blending (for example when the strings in the two languages obey different word order rules), a morphosyntactic (and prosodic) simplification takes place in the blending utterance, suggesting that two distinct complete functional layers with full tense and aspect information cannot be simultaneously projected.

## 4. Simultaneous utterance of two constituents within the same clause: Role Shift

In this section, we propose to analyze a type of Role Shift, a well-studied phenomenon that is systematically attested in sign languages, at the light of the co-utterance hypothesis. More precisely, we propose that Role Shift can result when the subject or the object is co-articulated with the verb. In the next section we summarize some findings about Role Shift and in Section 4.2, we present our proposal.

### 4.1 Main properties of Role Shift

Role shift is a very common strategy across sign languages by which the signer describes an event by assuming the perspective of a participant in that event instead of describing it as an external spectator (cf. Lillo-Martin 2012 and Steinbach 2021 for two exhaustive state-of-the-art papers). In fact, to confirm how the field of sign is split on these issues, the terminology to describe this phenomenon is not consensual, 'role shift' being the preferred label by researchers adopting what we called the 'unifying approach' and 'constructed action' being the preferred label in what we called the 'modal approach'. Finally, 'transfert personnel' is the term frequently used in research papers written in French.

This terminological split reflects in part the type of Role Shift phenomena that the researcher decides to focus on. In fact, there are two main uses of Role Shift. The first one occurs with propositional attitudes verbs, including *verba dicendi*, and is illustrated by the example in (14). (15 is its non Role Shift counterpart). In these sentences, the NP 'Jean' is attributed a specific locus in the neutral space and the anaphoric link between this NP and the subject of the verb KISS can be established in two ways. The latter may be expressed by a third person pronoun, which is a sign pointing to the locus previously associated to the NP 'Jean'. This is the strategy in (15), which resembles indirect discourse familiar from spoken languages. However, the anaphoric link may be established in a second way, namely the signer may shift into the role of Jean and signals this change of perspective by moving their body towards the locus in the neutral space associated to Jean or by directing their eye-gaze in that direction. This happens in (14) where this perspective shift forces the signer to refer to Jean by using a first-person pronoun. Role shift is indicated by a series of non-manual markings, that may change from language to language (and from predicate to predicate), but typically include body lean towards the locus associates to the character whose perspective is adopted, movement of the head towards that position, change in the direction of eye gaze, as well markers depicting the face of the reported signer.

$\overline{\phantom{IX-1 MARIE KISS}}^{\text{rs-i}}$

(14) JEAN$_i$ IX$_i$ SAY IX-1 MARIE KISS

'Jean said "I kissed Marie" '

(15) JEAN$_i$ IX$_i$ SAY IX-3$_i$ MARIE KISS

'Jean said that he kissed Marie'

Researchers in formal syntax and semantics have extensively worked on cases like (14), in which Role Shift is used to report utterances or thoughts of another person. We will follow Schlenker (2017) and call these cases 'Attitude Role Shift'. Attitude Role Shift occurs in bi-clausal contexts, since a propositional attitude verb is overtly expressed or must be understood.[2]

In fact, despite some important changes in the way the analysis is implemented, most formal analyses take the role-shifted clause to be embedded under a propositional attitude verb (Lillo-Martin 1995; Quer 2005; Schlenker 2017 a.o.) and explain that IX-1 in (14) refers to the signer in the reported speech act by postulating the presence of a context-shifting operator, namely an operator that can change the context relative to which an indexical is interpreted. This operator is taken to sit in a dedicated position in the periphery of the Role Shift clause. The lexical material uttered under Role Shift are the signs contained in the c-command domain

[2] Davidson (2015) proposes that Role Shift should be analyzed as a special case of direct quotation (cf. Maier 2018 for a further development and Quer 2013 that the claim that Attitude Role Shift needs not to be verbatim) similar to colloquial expressions introduced in English by "to be like" ("He was like 'I am being tired by you guys' "). Although Davidson's theory is appealing, the identification of direct quotation and Attitude Role Shift is not without challenge, as discussed by Schlenker (2017), who applies tests like extraction out of the role shifted clause and licensing of negative polarity items and shows that the role-shifted clause has properties typical of subordination. This suggests that Attitude Role Shift cannot be assimilated to direct quotation, which does not have these properties.

of the Role Shift operator. This domain is indicated by non-manual markings typical of Role Shift.

Yet, Role Shift is not restricted to propositional attitude verbs, as it is largely used to report non-linguistic actions of other people. (16), in which the signer shifts their body towards the position associated to Jean, is an example. By uttering (16) the signer re-acts Jean's gracious action, instead of describing it by adopting an external point of view.

____________________________rs-Jean

(16) JEAN$_a$ IX$_a$ ARRIVE HOUSE$_b$ IX$_b$ MARIE$_b$ FLOWER IX$_a$ CL(closed 5): 'donate_flower'

'Jean arrived home. He donated flowers to Marie.'

By uttering (16), the signer does not simply say that Jean gave flowers to Marie as a present but uses their body to show how this action has been performed (for example, the signer reproduces Jean's sweet facial expression). This type of Role Shift can occur in a mono-clausal context (a propositional attitude verb is not required). Again, following Schlenker (2017), we call the Role Shift illustrated in (16) 'Action Role Shift'.

Non-manual marking of Action and Attitude Role Shift are similar but cannot be fully identified. For example, body lean is often absent in Action Role Shift when the body posture and the facial expression of the impersonated character make evident that a change of perspective has occurred.

Action Role Shift seems to have a strong gestural component, and this is a challenge for researchers in the unifying approach, which is typically not well equipped to integrate that dimension. This is the reason why we focus here on Action Role Shift and try to extend to it the co-utterance hypothesis. Importantly, while the cases of co-utterance considered in that section involved co-articulation of entire clauses, we will claim that Action Role Shift involves co-articulation of two constituents inside the same clause, namely subject and verb (or object and verb).

We do not further discuss Attitude Role Shift, since we assume that the context-shifting operator approaches are on the right track (cf. Aristodemo et al. 2022 for experimental support to that general approach). The claim that Attitude Role Shift and Action Role Shift are two different phenomena requiring different analyses is further supported by an empirical observation that we review in the next section.

### 4.2 A co-utterance analysis of Action Role Shift

The co-utterance hypothesis allows us to offer a very simple analysis of Action Role Shift. Let us take as an example the matrix clause in (17): under this analysis, the clause displays a case of co-utterance of the subject (a pronoun pointing to the locus associated to the noun phrase JEAN, which in turn is the subject of the adverbial clause[3]) and of the verb DONATE. As a reflex of this co-articulation process, when the verb is articulated, the signer moves their body or points their eye-gaze toward the locus of JEAN and the sentence is interpreted from the perspective of Jean. Therefore, although the verb DONATE starts from the signer's body, it is interpreted as if JEAN were the agent of the donating action.

[3] That JEAN ARRIVE is an adverbial clause is indicated by the raised eyebrow non-manual-marking occurring over it. Cf. Wilbur 2012 for the claim that raised eyebrows is a common strategy to marks topics.

(17)

__________re                    __________rs-i

JEAN$_i$ ARRIVE   BOOK  IX$_i$  1-DONATE-2

'When Jean arrives, he will give you the book'[4]

We believe that there is a strong empirical argument in favor of the application of the co-utterance hypothesis to Action Role Shift. Take (18) as a representative.

(18)

______ RSa      ________re      _______RSb

STUDENT$_a$ TROUBLE.  IX$_b$ TEACHER$_b$   $_1$-HELP-$_a$

'A student had problems. The teacher helps the student'

As indicated by raised eyebrows, the sign TEACHER functions as a topic in the second utterance. The pointing sign IX, which precedes TEACHER, serves the purpose of giving the latter sign a locus in the signing space (TEACHER is articulated on the body, therefore a pointing sign is necessary to assign it a locus). The end point of the movement associated to the verb HELP is the locus of STUDENT in (18).

Under the co-utterance hypothesis, the verb is co-uttered with the pronoun IX referring to TEACHER, so the signer 'becomes' the teacher. The argument supporting the co-utterance hypothesis is the following: as by hypothesis a subject pronoun is structurally present and is co-uttered with the verb, the subject slot, being already occupied, should not be available for other categories. In turn, this means that, everything else being equal, inserting a subject pronoun IX$_1$ should not be possible. This prediction is correct. As originally observed by Engberg-Pedersen (1993, 1995) for DSL (and later replicated by Lillo-Martin and Quadros 2011 and by Schlenker 2017 for ASL), the presence of the subject pronoun IX$_1$ completely changes the interpretation of Action Role Shift sentences. This is shown by the contrast between (18) and (19). (19) contains an introductory part (indicated in parentheses for the reader's convenience) that reports that the teacher supervises the classroom. In this part Action Role Shift is already observed (the signer becomes the teacher). The expression 'and (s)he does what?' follows[5] and finally we come to the part which is of direct interest for us and is minimally different from (18): in (18) the verb HELP alone is under role shift while in (19) not only the verb but also the indexical IX$_1$ is.

(19)

______________________________RSa        ___re        ______ RSb  __________RSa

(TEACHER$_a$ **IX$_1$** SUPERVISING THE CLASSROOM . IX$_a$ WHAT) IX$_b$ STUDENT$_b$ TROUBLE.  **IX$_1$** $_1$-HELP-$_b$

'The teacher is supervising the classroom. And (s)he does what? (S)he says that (s)he helped the student who was in trouble'

As indicated by the translations (18) and (19) have quite different meanings. By uttering (19), the signer does not demonstrate what the teacher did but *says* what the teacher did, namely

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[4] Actually, depending on the context the sentence can also mean 'When Jean arrives, he will give him/her the book as a present'.

[5] Arguably this expression serves the function of facilitating the transition from the first part of the sentence where there is Action Role Shift to the second part where Attitude Role Shift is forced by the presence of IX$_1$.

the insertion of $\text{IX}_1$ transforms a case of Action Role Shift into a case of Attitude Role Shift. Why? Under the co-utterance hypothesis, we know why Action Role Shift is not allowed in (19): there are two competitors for one position, namely the structural position for the subject of HELP. As for the availability of Attitude Role Shift, the meaning of the sentence reveals that there is a silent verbum dicendi (a silent version of SAY). Therefore, (19) involves a biclausal structure: the clause with the silent SAY and the clause with the reported action. $\text{IX}_1$ is the subject of the silent verbum dicendi, not the subject of HELP. The version with the overt say is found in (20).

(20)

_____________________________RSa ___re ______ RSb __________RSa

(TEACHER$_a$ **IX$_1$** SUPERVISING THE CLASSROOM) IX$_b$ SAY STUDENT$_b$ TROUBLE. **IX$_1$** $_1$-HELP-$_b$

'The teacher is supervising the classroom. (S)he says that (s)he helped the student who was in trouble'

It is worth stressing that the co-utterance hypothesis introduces no generic ban against two categories referring to the same character. In fact, in sign languages it is very common to have multiple markers for the same argument via pronoun doubling, and LSF is no exception. The co-utterance hypothesis does not exclude the presence in the same sentence of two categories referring to the agent, as long as they occupy different positions, say the canonical subject position and a position specialized for Topic or Focus. This is shown by the sentence (21):

(21) JOHN$_i$ IX$_i$ EAT SANDWICH

(21) is acceptable because IX$_i$ occupies a Topic position in the right periphery. Therefore, IX$_i$ does not compete with JOHN for the canonical (preverbal) subject position. In fact, under the co-utterance hypothesis, even in (18) the agent TEACHER is expressed twice, first as a topic and second as the subject co-uttered with the verb. However, (19) is different because IX occupies the canonical subject position (Spec.TP), as indicated by the fact that the Role Shift non manual marking spreads over it. This position is already filled by the subject that is co-uttered with the verb and this creates a conflict that can only be fixed by switching to the Attitude Role Shift interpretation, which creates a second subject position.

Importantly, according to the co-utterance hypothesis in principle categories other than the subject might be co-uttered with the verb, if this is articulatorily possible. This applies to the (direct) object, as we now show. Indeed, cases of Action Role Shift in which the signer adopts the theme's perspective are attested alongside more well-known cases where the signer adopts the agent's perspective. For concreteness, let us call Subject Action Role Shift cases in which the signer shifts into the perspective of the Agent (like in 17 or 18) and Object Action Role Shift cases in which the signer shifts into the perspective of the Theme. (22) is a case of Object Action Role Shift. In (22) the end point of articulation of the verbs BEAT and SAVE IS the signer body, so the signer 'becomes' the student who is beaten and later saved (the sign STUDENT is signed on the body and is attributed a locus by eye gaze).

(22)

STUDENT WALK POLICE$_i$ $_i$-BEAT-$_1$ PEOPLE$_j$ $_j$SAVE$_1$

'A student was beaten by the police but was saved by (some) people'

The co-utterance hypothesis makes for Object Action Role Shift the same prediction that it makes for Subject Action Role Shift, namely, everything else being equal, it should not be possible to add an explicit *internal* argument since, by assumption, the internal argument slot is already occupied by the co-uttered object. This prediction is born out as well. This is shown in (23), which is identical to (23) but for the fact that the pointing sign IX$_1$ is added. As with the previous case, the sentence is grammatical, but a silent meaning verb SAY, which can host IX$_1$ in its subject position, is introduced and the sentence becomes a case of Attitude Role Shift.

(23)

STUDENT IX$_1$ WALK POLICE$_i$ $_i$-BEAT-$_1$ PEOPLE$_j$ $_j$SAVE$_1$

'A student said that he was beaten by the police but was saved by (some) people'

Finally, it is even possible to build a complex case in which the insertion of the pointing sign IX$_1$ transforms a sentence that contains both Subject Role Shift and Object Action Role Shift. This is (24):

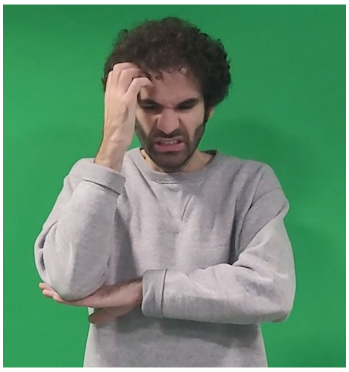
(24)

___________RSa ________RSa

STUDENT$_a$ IX$_1$ TROUBLE IX$_b$ PROFESSOR$_b$ $_b$-HELP-$_1$

'A student says that he is struggling and the teacher is helping him/her'

(24) contains two utterances. In the first one, the signer role shifts into the troubled student, as clearly indicated by this frame in the video:



However, the pointing sign IX$_1$ triggers the by now familiar insertion of a silent verb SAY, so (24) means 'the student said he is in trouble'. This is a case of *Subject* Action Role Shift transformation into Attitude Role Shift. (24) continues with an instance of *Object* Action Role Shift, as shown by the fact that the end point of articulation of HELP is the signer body, so the signer 'becomes' the theme of the action. Here as well there is a switch to Attitude Role Shift. The utterance conveys the information that the student *said* that he or she was helped by the professor (Attitude Role Shift with silent SAY).[6]

[6] Schenkler (2017) proposes that overt indexicals require a complete context (like the actual speech/sign act context), while the context introduced by Role Shift is too impoverished to support their interpretation. This explanation is not specific to pointing signs for persons and this might be an advantage if pointing signs in general (including temporal and locative pointing signs) trigger an attitude Role Shift interpretation, as Schenkler claims. On the other hand, there is a sense in which body movement or eye gaze towards a given locus is just another way to point to the locus and this continuity between IX and other ways of pointing is lost in that account, since only IX would require a complete context.

A reviewer asks whether the co-utterance hypothesis can be extended to structures that have a null pronoun but do not involve Role Shift. In fact, in many sign languages, including LSF, a subject or an object can remain unexpressed if its semantic content can be retrieved, like in (25) and (26):

(25a)

dominant: CL_PERS$_b$

IX$_a$ MARIE$_a$ IX$_b$ PIERRE$_b$

non-dominant: CL_PERS$_a$

'There are Pierre and Marie'

$_a$QUESTION$_b$ IX$_2$ CHOCOLATE LIKE ?

'Marie asks Pierre : 'Do you like chocolate ?''

IX$_1$ CHOCOLATE ADORE

'I love it'

(25b)

dominant: CL_PERS$_b$

IX$_a$ MARIE$_a$ IX$_b$ PIERRE$_b$

non-dominant: CL_PERS$_a$

'There are Pierre and Marie'

$_a$QUESTION$_b$ IX$_2$ CHOCOLATE LIKE ?

'Marie asks Pierre : 'Do you like chocolate ?''

ADORE

'I love it'

(26a)

dominant: CL_PERS$_b$

IX$_a$ MARIE$_a$ IX$_b$ PIERRE$_b$

non-dominant: CL_PERS$_a$

'There are Pierre and Marie'

IX$_a$ MARIE$_a$ $_a$QUESTION$_b$ IX$_c$ JEAN$_c$ CHOCOLATE LIKE ?

'Marie asks Pierre: Does Jean like chocolate?'

YES IX$_c$ CHOCOLATE ADORE

'Yes, he loves it'

(26b)

dominant: CL_PERS$_b$

IX$_a$ MARIE$_a$ IX$_b$ PIERRE$_b$

non-dominant: CL_PERS$_a$

‘There are Pierre and Marie’

$_a$QUESTION$_b$ IX$_c$ JEAN$_c$ CHOCOLATE LIKE ?

‘Marie asks Pierre: Does Jean like chocolate?’

YES ADORE

‘Yes, he loves it’

We do not propose an extension of the co-utterance hypothesis to these cases, though. First, we assume that co-utterance between a verb and its argument is overtly marked, for example by body movement towards the locus of the endorsed character or by eye-gaze in that direction. Although there have been attempts to link the presence of null subjects or objects to eye gaze or head tilt towards the locus of the missing argument (cf. Bahan et al. 2000 on ASL), experimental work does not support this hypothesis for ASL (Thompson et al. 2006). Similarly, the version of (25) and (26) with pronoun drop does not contain any non-manual pointing to the locus of the missing argument. Second, and more relevantly, we argued that the presence of IX in Role Shift triggers a systematic change of interpretation. This does not happen in cases like (25) and (26), as the versions with and without pronoun drop are largely synonymous. An analysis of null arguments in non Role Shift contexts is outside the scope of the paper, but we refer to Koulidobrova (2017) for discussion.

In this section we have defended the hypothesis that Action Role Shift is just co-utterance of the verb with the External or with Internal argument and that the presence of a pointing sign triggers a change of interpretation that is necessary to make space to this element.

## 5. Incorporation as opposed to the simultaneous utterance

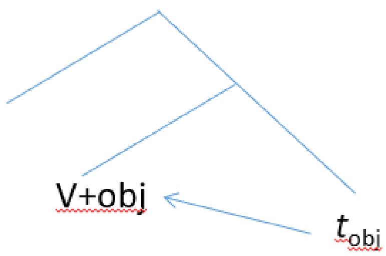
It is important to distinguish the co-utterance hypothesis from the morphosyntactic process of incorporation. In many sign languages, the verb can be modified to indicate the size and shape of the internal argument. For example, in the pictures in (27), the special way the LIS verb for ‘lift’ is articulated indicates that the theme is a heavy box. Therefore, it is tempting to look at cases like (27) as noun incorporation cases, by which the noun stem combines with a verb and yields a more complex predicate (cf. Massam 2009 for an overview on noun incorporation in spoken languages).

(27)

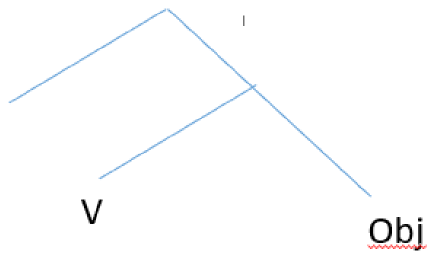
Under the incorporation analysis, a single morpho-syntactic unit, the verb, expresses the information that can be widespread through the clause (as in polysynthetic languages). This type of account would make sign languages akin to polysynthetic languages but for the fact that the information conflated on the verb is realized simultaneously in sign languages while it is realized sequentially in polysynthetic languages (morpheme after morpheme). However, in this section we argue that an incorporation analysis is not viable.

There are two main ways to conceptualize noun incorporation. According to a first approach, opened by the seminal work of Baker (1988), incorporation is a syntactic operation (head movement) with a morphological reflex. According to other approaches, like the ones by Rosen (1989) and Di Sciullo and Williams (1987), the operation combining the verb and the internal argument would take place in the lexicon. Although these approaches differ for the postulated timing of the formation of the complex predicate (lexicon or syntax), they both assume that. by the moment the hierarchical structure is linearized, the combination of the verb and of the incorporated category is already in place. This is exactly where the co-utterance hypothesis differs from an incorporation analysis. The gist of the co-utterance hypothesis is that two categories *that are independent* can be uttered at the same time if this is articulatory possible. Assuming Baker's approach for concreteness, the structure underlying noun incorporation is schematically illustrated in (28) while the structure underlying simultaneous utterance of object and verb is illustrated in (29).

(28)




(29)




In order to show that incorporation is not a viable option for the facts under discussion in this paper, we capitalize on a solid cross-linguistic observation, namely that incorporation is prototypically limited to internal arguments, while agent incorporation is very rare. In fact, to the best of our knowledge, putative cases of agent incorporation have been described only in Turkish (Öztürk 2009[7]) and Indonesian (Myhill 1988) while theme incorporation is productive in languages in many areas of the world (cf. Gerdts 2017). If incorporation results from a

[7] In fact, Öztürk (2009) considers this hypothesis for Turkish but ultimately rejects it because she concludes that agent incorporation cannot be equated to genuine cases of noun incorporation to begin with.

syntactic operation (head movement), it is easy to explain why agent incorporation is rare: in the simplest scenario, agent incorporation results in the violation of the condition that dictates that a trace must be c-commanded by its landing site. As for exceptions (Turkish and Indonesian), although a detailed analysis of these cases is outside the scope of this paper, one can note that agent incorporation may obey the c-command requirement if it occurs after the verb head raises to a position where it can c-command the agent, e.g. higher than T. The rarity of agent incorporation might then be due to the fact this implies a derivation much more complex than simple head movement of the incorporating category. Therefore, to exclude the incorporation analysis for what we claimed to be simultaneity cases, we need to exclude the derivation where the agent allegedly incorporates into the verb after the latter has moved to a higher functional projection than the former.

Let us consider this hypothetical incorporation analysis for Role Shift by taking a concrete example. (30) is a biclausal structure in which the direct object of each clause ('PEOPLE WEAK' and 'PEOPLE STRONG' respectively) is topicalized (the presence of a subtle nonmanual marking for topic in the first clause has been confirmed by our consultants while the topic in the second sentence, which is more evidently marked by non-manuals, is contrastive in the sense of Frascarelli and Hinterhölzl 2007). Assuming that topic categories sit in a low position in the left periphery (Rizzi 1997), the word order in (30) suggests that the verb has not accessed the left periphery of the clause and sits in its canonical position (say, T). Therefore, if Action Role Shift were incorporation of the subject into the verb, (30) would be a violation of c-command requirements.[8]

(30)

JEAN$_i$ IX$_1$ HATE.

                    ________________RC-i                    ____________RC-i

IX$_i$ PEOPLE WEAK IX$_i$ LOOK-WITH-CONTEMPT,    PEOPLE STRONG IX$_i$ LOOK-WITH-FEAR

'I hate John. He despises weak people but he fears strong ones'

Having established that incorporation is not a viable analysis for Role Shift cases involving body shift and/or eye gaze and that co-utterance is independently motivated, let us go back to cases traditionally analyzed as incorporation like (27) (or similar ones, like the signs for eating a sandwich or an apple, which reproduce the shape of the ingested object). As the c-command requirement is respected in these cases (the trace of the incorporating internal argument is c-commanded by the verb complex), an incorporation analysis is possible in principle. Still, as the co-utterance hypothesis can extend to these cases, Occam's razor considerations lead us to dispense with incorporation (at least in syntax[9]). While assuming the co-articulation analysis, we note that in prototypical cases of Role Shift, the articulator which is involved in co-utterance is the entire body of the signer (through body movement or through eye gaze).

[8] A reviewer asks what blocks a putative derivation in which *v*-to-T precedes Spec,*v*P to Spec,TP and the subject incorporates into the complex v/T head starting from Spec,*v*P. This derivation is conceivable, but it requires two distinct mechanisms of subject licensing, the canonical Sped-Head configuration between Spec,T and T (when there is no Role Shift and there is an agreement verb) and licensing by incorporation (when there is Role Shift). On the other hand, the co-utterance hypothesis does not need to postulate two distinct licensing mechanisms, as Role Shift is assumed to be co-utterance starting from the canonical subject position in Spec.T.

[9] Santoro (2018) offers a systematic description of compounds in LSF and LIS including simultaneous compounds.

However, in incorporation-like cases, the articulators are body parts, i.e. the mouth and the hand(s) reproducing the shape of the ingested object in signs representing eating a sandwich or an apple. This may have an effect if, as proposed by Meir et al. (2007), the (entire) body of the signer has a privileged status because it encodes the argument which the verb is predicated of, that is, it is the subject of predication.

In this section we argued that classical cases of Role Shift cannot be reduced to cases of incorporation, and we considered a co-utterance analysis of cases traditionally analyzed as incorporation.

## 6. General considerations about linearization of hierarchical structures

In this paper, we proposed that two (or more) clausal constituents can be co-uttered under certain conditions and that two constituents inside the same clause can be co-uttered. As for the co-utterance inside the clause, we extensively argued for the case of co-utterance of subject and verb (Subject Action Role Shift) or of object and verb (Object Action Role Shift). A natural reaction when confronted with this proposal is that, if it is not constrained, it can lead to massive overgeneration.

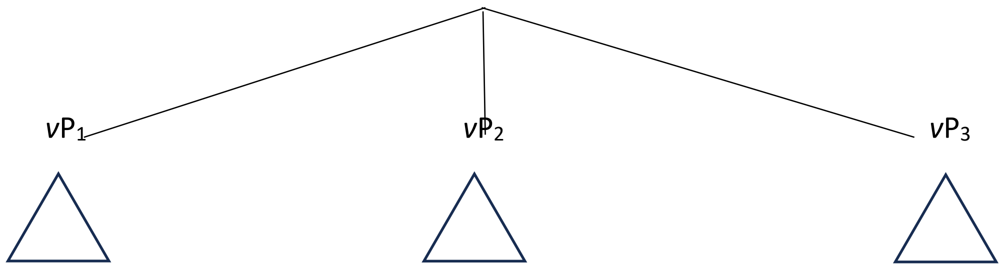
We discussed some limitations concerning co-utterance of clausal constituents in Section 3, where we argued that these constituents, which deliver different aspects of the same to-be conveyed message, are coordinated *v*Ps sitting under a unique functional layer.

There are other (structural) constraints, as suggested by the fact that many cases of co-articulation are not attested. For example, the co-utterance of subject and object while the verb is articulated on its own does not seem to be attested and it is even difficult to imagine what a similar structure would look like. In order to tackle the question of the structural constraints on co-uttered constituents, consider that co-utterance can be seen as a way to suspend the linearization of the hierarchical structure, namely the operation that flattens the syntactic tree into a linear sequence. For this reason, developing a full theory of co-utterance would require developing a full theory of linearization, a point that exceeds the limits of this paper. However, there is a hint that suggests the format that a restrictive theory of co-utterance should have and this is the observation that in all relevant cases of simultaneity the linguistic categories that are co-uttered do not need to be so (simultaneity is an option, not an obligation). Furthermore, a descriptive generalization seems to emerge: the configuration that allows co-uttering is mutual c-command[10]. Let us see why it is so by considering some representative cases of co-uttering. First, a verb and a direct object c-command each other in the VP and this is compatible with a co-uttering analysis of apparent incorporation cases like (27). Second, the subject and the node T' c-command each other and this the basis of Subject Action Role Shift cases like (23). Third, Object Action Role Shift cases like (24) are captured by the mutual c-command requirement condition if the object IX and the verb mutually c-command in the VP. Fourth, the mutual c-command requirement can account for highly simultaneous structures like (1), at least if it assumed that these types of coordinated constituents are symmetric (cf. Neeleman, et al. 2023 for a defense of the idea that coordinated constituents can be symmetric). Under the analysis illustrated in (31), the *v*Ps that sit under the shared functional layer do c-command each other.[11]

[10] We thank an anonymous reviewer for this suggestion.

[11] A symmetric analysis of coordination is not necessary if it is assumed that the condition on co-utterance can be recursive, namely $\alpha$ and $\beta$ can be co-uttered if they c-command each other and $\{\alpha,\beta\}$ can be co-uttered with $\gamma$ if $\{\alpha,\beta\}$ and $\gamma$ c-command each other. This recursive rule predicts that under certain conditions an entire

(31)




‘there is a fence’ ‘the horse looks at *it*’ ‘the bird is watching *him*’

Other cases of simultaneity discussed in the literature might fall in place as well. We refer to the fact that in many sign languages, a manner adverb can appear as an independent sign or, alternatively, the adverbial meaning is expressed by modifying how the verb is articulated. (32) shows the first option: QUICKLY is signed as a separate lexical item. (33) shows the second option: QUICKLY forms a unique sign with the verb and, as a consequence, the movement of the dominant hand towards the mouth of the signer, characteristic of the sign EAT, is repeated and is articulated more rapidly than in the citation form of the verb. In this case as well, if a manner adverb is adjoined to the VP, one can say that these two categories c-command each other and co-uttering is possible.

(32) MARIE$_i$ MEAT IX$_i$ EAT QUICKLY

‘Maria eats meat quickly’

(33) MARE$_i$ MEAT IX$_i$ EAT-QUICKLY

‘Maria eats meat quickly’

Still another candidate to be analyzed as a case of co-utterance is negation when this is expressed by a nonmanual marking which is simultaneous with the negated constituents.

Importantly, the mutual command hypothesis excludes many theoretical cases of simultaneity. We already mentioned the case of co-utterance of subject and object excluding the verb. Another impossible case involves a temporal and a locative adjunct: even if they are linearly adjacent, they are expected *not* be co-utterable, if an adverbial is adjoined to the portion of the clausal skeleton it modifies, say the VP. An adverb is never a sister node of another adverb, so the mutual command hypothesis excludes that they be co-uttered.

Whatever the exact formulation of the general condition that allows co-uttering, mutual c-command or another one to be refined in further research, a general consideration is suggested by the facts that we have investigated. In current generative approaches, there are two main ways to conceive linearization. The first one goes back to Kayne’s (1994) Linear Corresponding Axiom and its central tenet can be summarized by saying that asymmetric c-command imposes a linear ordering of terminal nodes. More specifically, if $\alpha$ c-commands $\beta$ and $\beta$ does not c-command $\alpha$, $\alpha$ is bound to precede $\beta$ when the syntactic tree is flattened into a linear sequence. The crucial observation for us is that in Kayne’s theory, hierarchical structures inherently encode linearization, in the sense that a tree comes equipped with

sentence can be co-uttered. A possible candidate for this global co-utterance can be predicates involving a handling classifier, in which a single sign encompasses agent, theme and verbal root. We leave this question for future research.

information of what will precede what, since a tree is defined by (lack of) c-command configurations between its nodes.

A different view about linearization has been proposed by Chomsky, starting from Chomsky (1995) until the implementation in Chomsky (2019): the idea is that core syntactic structures are unordered, as they result from the operation Merge, which is a set-creation operation. Order is imposed only when syntactic structures are shipped to the phonological component, because at this stage linearization is forced. Although asymmetric c-command may still play a role, its role is encoded in a linearization algorithm that takes as an input a structure that by itself is unordered.

In this paper, we argued that linearization can be suspended when articulatory constraints allows this. This suggests that linearization is not *automatically* determined by asymmetric c-command, as there are trees that remain (partially) unordered. For example, if we are right, in Subject Action Role Shift, the subject NP asymmetrically c-commands the verb, still the two are not linearized (as they are co-uttered, they are not ordered with respect to each other).

We conclude that the co-utterance hypothesis is more consistent with Chomsky's claim that core syntax is pure hierarchy (with no intrinsic order).[12]

In our discussion we have been conservative in one respect, namely by arguing that linearization can be *suspended*, we implicitly assumed that linearization is the default option. In principle, one might go the opposite way and argue that language is inherently multidimensional and linearization is the "special case", being forced only because each spoken word must be pronounced after the other. Under this view, linearization would be an accident of the phono-articulatory modality. One of the reasons that lead us to think that linearization is not the special case is that sign languages make a full use of the three-dimensional space only in certain cases, while in other cases the three-dimensional space is not exploited even when in principle it could be. In fact, simultaneity is always an option, never an obligation. Even more telling, in reciprocal constructions, as described in Section 3.1, simultaneity is not forced, although the event described by the sentence is by definition simultaneous. In principle, data from acquisition are relevant to this question. If lack of linearization (namely presence of multidimensionality) were the default option for natural languages, deaf signing children should start out with simultaneous structures and should switch to sequential utterances only later in their acquisition path. Although the evidence on acquisition of sequential as opposed to simultaneous structures is fairly limited, the available evidence shows that simultaneous structures, at least complex ones, arrive later (cf. Sallandre 2004).

Finally, in principle our proposal does not exclude cases of co-utterance involving spoken language when articulatory constraints allow this, say when a person chats on WhatsApp and talks at the same time. However, the limits we impose on co-utterance (mutual command and forced origin from the same Conceptualizer level) severely constraint this possibility.

[12] An alternative proposed by a reviewer is that a fully linearized structure in the syntactic component could be partially delinearized at the PF component to allow (some) co-utterance.

## 7. Summary and conclusion

Our starting point in this paper was the observation that proficient bimodal bilinguals can simultaneously utter a sentence in a sign language and the corresponding one in a spoken language. As spoken and sign languages use largely independent articulators, bimodal production is an ideal case to study the simultaneous activation of grammatical systems. However, sign languages also use multiple articulators (the two hands, the facial expressions, the eye gaze, the position of the torso etc.) which work in coordination but have a degree of autonomy. So, we proposed that the simultaneous production of linguistic categories can take place internally to a single sign language as well.

We explored this possibility and argued that it is possible to co-utter two or more coordinated clauses, if they contribute to convey the same complex conceptual message. This mechanism is at origin of many utterances that transmits a global picture of a complex event. Different, although simultaneously expressed, clausal units can describe a unique event because they are linked by spatial anaphoric relations. We can now provide an answer to the initial challenge that motivated this article, which was providing a syntactic analysis to massive cases of simultaneity like (1). If it is possible to co-utter clausal units, a case like (1), no longer presents special analytical difficulties. In this example, three distinct propositional contents are co-uttered:

(i) 'there is a fence'

(ii) 'the horse looks at *it*'

(iii) 'the bird is watching *him*'

The question of how to identify the syntactic structure if (1) is easy to answer, as for each of the three units it is relatively straightforward to identify a subject and a predicate. As for the fact that in (1) the antecedents of the pronoun is determined uniquely unlike what happens in the coordination of (i), (ii) and (iii) in English, we explained this with the specific properties of spatial anaphora.

We also argued that it is possible to co-utter two constituents inside the same clause. This is what happens in Action Role Shift, for example.

We do acknowledge that the co-utterance hypothesis is still largely a working hypothesis which should be further developed in future work. Still, even at the present stage, we think that it has merits that warrants this exploration, even more so because it makes some specific predictions about the phenomena under analysis that allow to assess it on empirical ground.

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